Introduction

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Drug discovery is an arduous journey with many unknowns throughout the path that may lead to failures. To perform a large number of trials iteratively, computer simulations are convenient enabling tens of thousands of calculations. This sheer number of calculations can provide crucial insights, and possibly, rule out dead ends very early in the discovery cycle. Drug discovery is known to be a multi-faceted science that requires a great deal of art and innovative thinking, and often luck plays a good role. For the latter factor, we have no control, however, the best we can do is to base our science on solid foundation. In computational chemistry and biology, we often aske an important question- how do we train computers to think like humans? However, this is not a worry because we do not need to train computers to think like humans. As a matter of fact, computer are extremely fast but dumb machines. Whereas humans are extremely slow, lazy yet smart thinkers. Now, the division of labour between humans and computers should be straightforward; let the computers do the repetitive tasks of solving several complex mathematical equations that can be programmed to describe chemistry and biology. This is done by employing the laws of physics that govern the movement of the atoms, and therefore molecules. Humans can focus on the complex decision-making steps that are beyond mathematical equations. However, with generative artificial intelligence (**GenAI**), computers can perform some level of thinking. However, their chemical and biological validity, in a general sense, remains to be tested. GenAI, therefore, deserves an appreciation to at least recognise the idea that computers could be made to think, not in a biological sense, rather in a mathematical sense, which can be converted to chemical or biological models. However, the latter part has been extremely challenging and yet to be conquered entirely.

The aim of this book is to ensure that the reader, perhaps, a beginner is aware of many assumptions that enable quick calculations. Also, thoroughly understand the limitations of these methods and to use them to perform calculations that they are meant or developed for. There are certain critical assumptions that ensure the speed of the calculations are, neglecting the solvation effect, and neglecting the entropy part of the free energy of binding. These, and more limitations will discussed in great detail in this later chapters of this book. It must be noted that such assumptions are somewhat important to ensure that the calculations are fast enough to finish several hundreds or thousands docking calculations per day, as warranted to virtually screen huge databases that comprise few million to billion molecules.

This chapter can be treated as a short summary on topics that will be covered in great detail in the following chapters. Note that this book is intended to be a practical guide, therefore, we will emphasise more the practical aspects of molecular docking one needs to know before employing it to generate lead ideas or hit finding. We have made efforts to cover the theoretical foundations on topics that we deem crucial to successfully apply molecular docking. Depsite effects, we do not claim this book to be comprehensive text to cover all theoretical foundations on molecular docking. We further encourage readers to start practicing molecular docking, even though one may think they know nothing on this topic. We claim, with some personal experience on learning several computationals tools, the most learning happenings by trying first, and then consulting books like this one to sharpen your understanding.

1 Molecular Recognition and Binding

Understanding protein-ligand interactions at the atomic or molecular level is central to studying how small molecules interact with their biological receptor(s). Thus binding phenomenon may involve, but not limited to, multiple types of non-covalent interactions such as hydrogen bonds, electrostatic interactions, van der Waals forces, and hydrophobic interactions. The binding affinity between a ligand and its receptor can be determined by the sum of favourable and unfavourable interactions, with the overall binding affinity dictating the strength of the complex. Protein-ligand binding affinity prediction quantifies the binding strength, providing crucial insights for lead optimisation by changing functional groups that lead to maximum gains[16].

The binding affinity is determined experimentally in the form of inhibition constants (K_i), dissociation constants (K_d), or half-maximal inhibitory concentrations (IC_{50}). If one is using isothermal calorimetry, then direct measurements to thermodynamic qualities like the change in enthalpy due to ligand binding (ΔH), and the change in entropy ($T\Delta S$) can be made. However, experimental methods are expensive and time-consuming, thus, efficient computational methods are capable of making accurate protein-ligand binding affinity predictions are time and cost saving methods. Such accurate and high-quality computational methods can reduce the number expensive experiments by filtering unwanted molecules. The downside of computational methods are that they are heavily dependent on the empirical parameters used to make computing tractable.

2 Historical Background

2.1 Early Foundations

The conceptual foundations of structure-based drug design can be traced back to the early 20th century with Emil Fischer's lock-and-key theory proposed in the late 1800s[4]. This theory postulated that ligands fit into receptors like a key into a lock, representing the first mechanistic understanding of molecular recognition. This concept was later refined by Daniel Koshland's induced-fit theory in 1958[10], which recognized that proteins are dynamic in nature that undergo conformational changes upon ligand binding. The induced-fit theory suggested that the binding site of the protein is continually reshaped by interactions with the ligands as they interact with the protein, providing a more accurate description of binding events than the rigid treatment.

2.2 The Birth of the Protein Data Bank

A pivotal moment in the history of SBDD came in 1971 with the establishment of the Protein Data Bank (PDB) as the first open-access, digital-data resource in biology[2]. The PDB was created to gather structural biologists at a single platform and has evolved from being a collaboratory network to a full-fledged database with an extensive list of protein structures, nucleic acid-containing structures, and computational models. The availability of three-dimensional atomic level structures of biomacromolecules determined using X-ray crystallography, nuclear magnetic resonance (NMR) spectroscopy, and electron microscopy fundamentally transformed drug discovery.

2.3 Structural biology Revolution

Structural biology has undergone tremendous improvements over the past few decades that led to a direct impact on structure-based drug design (SBDD)[1]. X-ray crystallography provided the first atomic-resolution views of protein–ligand complexes and established the modern SBDD. High-resolution crystal structures give crucial information on binding pockets, hydrogen-bond networks, and water-mediated contacts. This information improves our understanding about the drug-receptor interactions with atomic details. Over time, improvements in crystallization methods, synchrotron sources, and detectors increased throughput and resolution, making co-crystal structures a routine decision-making tool in medicinal chemistry. Furthermore, NMR spectroscopy extended this static picture representation of proteins by resolving their stucture in solution under near-physiological conditions[1]. Most recent advancement has been in the field cryogenic electron microscopy (Cryo-EM). Improvements in direct electron detectors, image processing algorithms, and microscope stability has enabled near atomic level resolution for large complexes and membrane proteins that were previously intractable to crystallography or NMR. Single particle cryo-EM routinely yields structures of GPCR–G protein complexes, ion channels, transporters, and multi-subunit assemblies in multiple functional states, providing unprecedented insights into conformational landscapes and ligand-induced transitions. This has transformed SBDD for membrane proteins: structures of clinically important targets such as TRP channels, CFTR, and GABA receptors have revealed binding sites, gating mechanisms, and off-target liability determinants that now guide design of safer and more selective small molecules[17].

2.4 Evolution of molecular docking methods

Molecular docking has evolved from a geometric fitting algorithm on rigid receptor and ligands to algorithms that can handle flexibility in ligand and receptor simultaneously. The first widely recognized docking approach was developed by Kuntz and co-workers in 1982, implemented in the program DOCK.[12] This method treated both receptor and ligand as rigid bodies, and emphasized geometric complementarity: ligands were positioned in binding sites by matching molecular shape using spheres and surface representations. This "first generation method" established the conceptual framework of pose prediction and ranking, however neglected conformational flexibility and detailed energetics due to computational limitations. In the 1990s and early 2000s, many groups followed with major improvements to increase the accuracy of the docking method. Search algorithms such as systematic search, Monte Carlo based search, and genetic algorithms were employed to enhance the ligand sampling to find the bioactivity conformation. Since accuracy of the docking methods also depends on the scoring functions, empirical and force field-based scoring functions were developed to approximate binding affinity.[9, 21] Programs such as AutoDock incorporated a Lamarckian genetic algorithm and grid-based energy evaluation, that allowed a routine docking of ligands with many rotatable bonds against rigid receptors.[14] As the speed to docking ligands increased, this allowed for docking-based virtual screening of large chemical databases for hit identification, which significantly improved the productivity of early phase drug discovery.

The next logical improvement focused on adding receptor flexibility to account for induced-fit effect to mimic physiogical binding phenonmenon. However, the additional degree of freedom came a the cost of computational speed[21]. This was partially remedied by ensemble docking, which means instead of docking with receptor flexibility, docking was carried out on several conformations of the proteins that might have reported experimentally or with short runs of moecular dynamics simulations. Another milestone in the development of molecular docking methods was the ability to address covalent docking. Several of the drugs in the oncology section show their anticancer properties by covalently binding to its target. This was possible with algorithms capable of sampling non-covalent pre-reactive ligand topologies, followed by refining poses using post-reactive topologies that include the covalent bond[23, 20].

Recent developments focused on leveraging the machine learning–based scoring functions, explicit treatment of structural water, and dynamic docking approaches that couple molecular dynamics with docking searches[15]. Taken together, these improvements over several decades have significantly improved both computational speed and accuracy. Now a days, screening a large libray of millions to billion molecules is tractable. This can be equally attributed to parallel improvements in docking algorithms and computing power of modern day computers or supercomputers.

3 Structure-based drug design

Structure-based drug design (SBDD) uses the knowledge of the receptor structure and its binding site to understand the shape of ligands and the type of interactions required for binding. The lock-and-key hypothesis is the main assumption for developing software tools for SBDD. Therefore, the quality of the results depend on two things assuming the software used is accurate (more on this in later chapters explaining why this is not always true):

- Quality of the receptor structure
- Confidence with which the binding site information is known

The objective of molecular docking is to enables us to predict the binding modes, and thus, affinities of small molecules to their target receptor. It models the interaction between the atoms of small molecule and the receptor. Generally speaking, molecular docking is possible between any two (or more) chemical or biological entities, such as protein-protein, protein-DNA, protein-RNA and all such combinations that are biologically observed. All aforementioned binding and recognition follows the same rules of interactions governed by laws of physics. However, computationally, different potential energy functions are needed to model these binding. We mainly focus on small molecule protein docking, nevertheless, these discussion should be applicable to other kind of docking.

3.1 Principles of Structure-Based Drug Design

Structure-based drug design is fundamentally dependent on knowledge of the three-dimensional structure of the biological target. The iterative process of SBDD begins with the selection of a potential target, followed by ligand identification. The target may be a macromolecular protein involved in biosynthetic pathways. The three dimensional structure of the protein is developed using sophisticated techniques including X-ray crystallography, nuclear magnetic resonance spectroscopy, and cryo-electron microscopy. Once the structure of the target is identified, it is necessary to investigate the properties of the protein such as presence of cavities, clefts, amino acid sequences, and allosteric pockets. Using various available computer algorithms, drug-like molecules from large databases of small molecules or fragments of compounds can be docked into the binding cavities of the target protein.

3.2 The SBDD Workflow

Figure 1 shows a typical workflow for computational drug discovery. Depending on the chemical and biological information available, one might proceed by the structure-based approach or ligand-based.

3.2.1 Druggability and Target Selection

Druggability refers to the likelihood that a biological target can be modulated by a small molecule or any biological entity. The first step in SBDD is the choice of an appropriate target, followed by evaluation of the structure of that target. Targets must possess characteristics that make them amenable to small molecule binding, including accessible binding sites with suitable geometry and chemical properties.

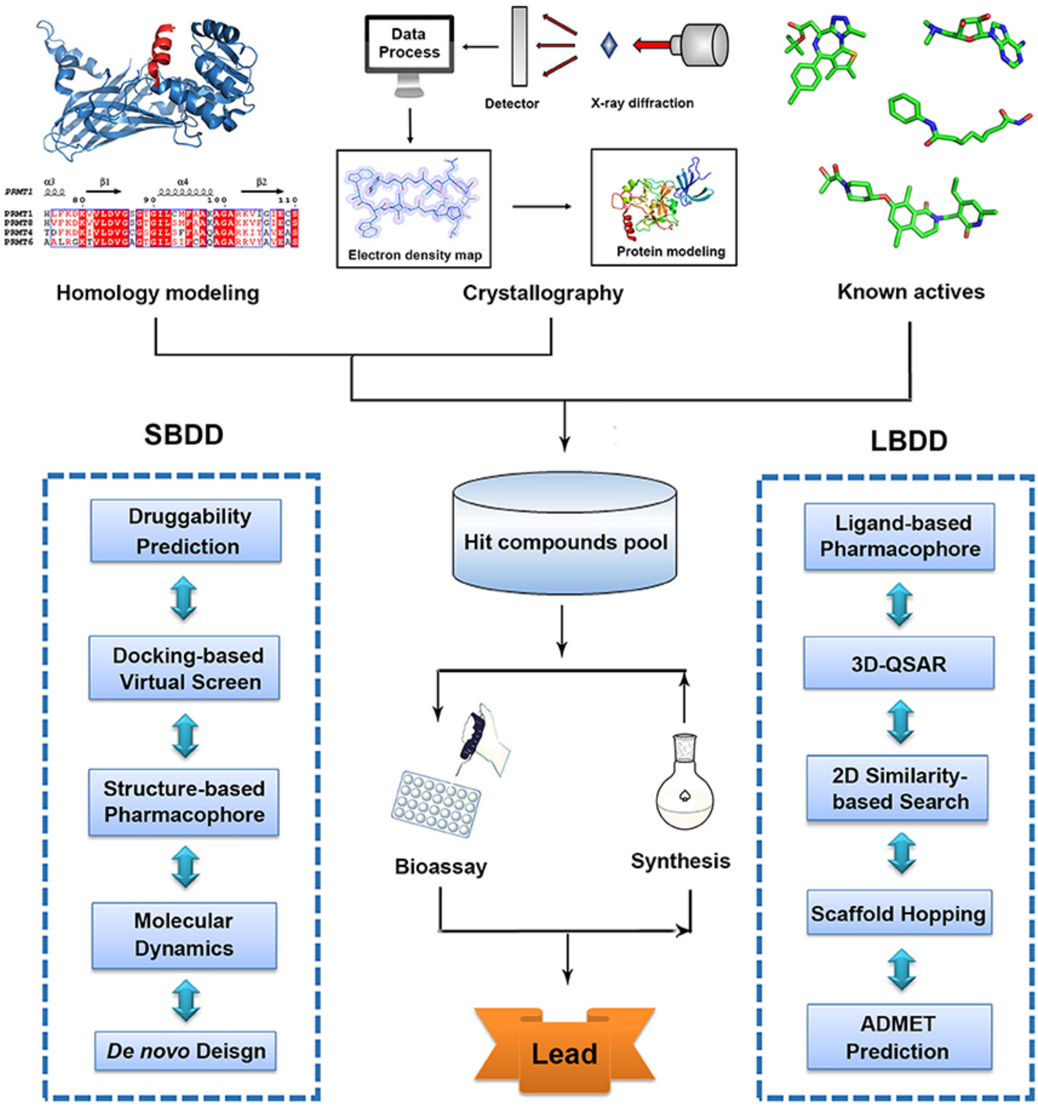

Figure 1: A typical computational workflow for drug design. This figure also compares an alternative workflow called ligand-based drug design (LBDD), which is used when receptor information is not available. This image is reproduced from[13] under the Creative Commons Attribution License (CC BY) licence.

3.3 Binding Site Identification

Knowing the location of the binding site before docking processes significantly increases docking efficiency. In many cases, the binding site is known before docking ligands into it, and information about the sites can be obtained by comparison of the target protein with a family of proteins sharing similar function or with proteins co-crystallized with other ligands. In the absence of knowledge about the binding sites, cavity detection programs or online servers such as GRID[6], SurfNet[6] and many more can be utilized to identify putative active sites within proteins. Docking without any assumption about the binding site is called blind docking. This could also be a strategy to explore possible sites that could binding cavities not identified before.

3.3.1 Determination and Preparation of receptor

The SBDD workflow begins with obtaining the three-dimensional structure of the target receptor. Structures can be obtained from the Protein Data Bank, which currently holds over 155,000 atomic level 3D structures of biomolecules. When experimental structures are not available, protein modeling including homology modeling and *ab initio* techniques allows for predicting the three-dimensional structure. Selection of appropriate receptor structure from the PDB is a critical step, as multiple crystal structures are often available for the same receptor. A systematic approach for selecting the right type of receptor structure must consider factors such as resolution, completeness, presence of ligands, and biologically relevant conformations. Recent advances include AlphaFold[8], which has revolutionized life sciences by predicting over 200 million protein structures available in the AlphaFold Database, far exceeding the number of experimentally resolved structures.

3.3.2 Ligand Library Preparation

After target structure preparation, the next step involves preparing libraries of compounds for virtual screening. Ligand preparation includes generating three dimensional coordinates, assigning proper protonation states, and optimizing geometries using suitable force field parametes. Compound libraries can range from commercially available compounds to virtual libraries representing chemical space that could be synthesized.

3.3.3 Virtual Screening

Virtual screening is a computational technique used to search large libraries of compounds and identify those most likely to bind to a drug target. Structure-based virtual screening uses molecular docking to test large libraries of compounds for potential binding to target proteins. The virtual screening process typically involves multiple stages of filtering and refinement. Initial rapid screening of large compound collections using fast scoring methods is followed by more detailed analysis of top ranked compounds using more sophisticated scoring approaches. Consensus virtual screening between multiple docking programs can increase reliability of results. Top scoring molecules from virtual screening with high affinity toward target protein are synthesized and tested in *in vitro* biochemical assays. Ligands with desired therapeutic activities are further evaluated for efficacy, affinity, and potency studies. Structural insights into the complex of ligand-protein assist the analysis of variety of binding conformations and interactions.

4 Molecular Docking: Concepts and Methods

In this introductory chapter, we attempt cover, albeit as brief note, all topics that will be covered in great detail in further chapters.

4.1 Fundamentals of Molecular Docking

Molecular docking is a computational technique that predicts the binding affinity of ligands to receptor proteins. The docking process involves two basic steps: prediction of the ligand conformation (referred to as pose) as well as its position and orientation within binding sites, and assessment of the binding affinity. These two steps are related to sampling methods and scoring functions, respectively. The scoring function guides and determines the most plausible poses from thousands of possible poses. It can be used to determine the binding mode and site of a ligand, predict binding affinity, and identify potential drug leads for a given protein target. Despite intensive research over the years, accurate and rapid prediction of protein-ligand interactions remains a challenge in molecular docking.

4.2 Rigid vs. Flexible Docking

The treatment of molecular flexibility represents a fundamental consideration in docking methodology. Early docking methods treated both ligand and receptor as rigid bodies based on the lock-and-key theory. Docking has been performed with a flexible ligand and a rigid receptor for a long time and remains the most popular method in use. This approach balances computational efficiency with reasonable accuracy for many applications.Considering the limitation of computer resources, flexible ligand docking with rigid receptors dominated the field, though recently many efforts have been made to deal with the flexibility of the receptor. Flexible receptor docking, especially backbone flexibility in receptors, still presents a major challenge for available docking methods. The induced-fit theory suggests that ligand and receptor should be treated as flexible during docking, as this could describe binding events more accurately than rigid treatment[11].

4.3 Docking Algorithms

Multiple algorithmic approaches have been developed for conformational sampling in molecular docking. Common search algorithms include genetic algorithms, Monte Carlo methods, simulated annealing, and systematic search approaches. The Lamarckian Genetic Algorithm implemented in AutoDock4[14] successfully modeled flexible ligand binding, particularly useful in complex systems requiring in-depth torsional and conformational exploration. AutoDock Vina[22] introduced a multithreaded architecture that outperformed AutoDock4 by up to 100 times in docking throughput. Using benchmark datasets, Vina correctly placed native-like poses in the top three results in more than 93% of situations, particularly excelling in polar and charged receptor settings. The Attracting Cavities algorithm represents another important approach, positioning cloud points in protein cavities and using these as guides for ligand placement and optimization.

4.4 Scoring Functions

4.4.1 Role and Importance

Scoring functions are mathematical functions[7] used to approximately predict the binding affinity between two molecules following molecular docking. During the docking process, the search algorithm investigates a vast amount of conformations for each molecule, and scoring functions evaluate the quality of these docking poses, guiding the search methods toward relevant ligand conformations. The scoring function calculates a score or an estimate of binding affinity of a particular pose. This represents the thermodynamics of interaction of the protein-ligand system, to distinguish true binding modes from all others explored and rank them accordingly. Scoring functions must accomplish three primary tasks. The first goal is the ability to distinguish experimentally observed binding modes, associating them with the lowest binding energies of the energy landscape from all other poses (pose prediction). The second goal is to classify active and inactive compounds during virtual screening. And the third is prediction of the binding affinity, ranking compounds correctly according to their potency (binding affinity prediction), which is the most challenging task. An ideal scoring function would be able to perform all three tasks effectively, however, given several limitations of current scoring functions, they exhibit different accuracies on distinct tasks due to modeling assumptions and simplifications made during their development phase.

5 Current challenges and limitations of molecular docking

5.1 Limited accuracy

Despite considerable success, accurate and rapid prediction of protein-ligand interactions remains a challenge in molecular docking. One of the biggest challenges is modeling the receptor flexibility to fully take in to account the induced-fit behaviour. Most molecular docking tools provide high flexibility to the ligand, while the protein is kept more or less fixed or provided with limited flexibility to residues present within or near the active site. Providing complete molecular flexibility to the protein increases the space and time complexity of computation exponentially. The accuracy of the scoring function to estimate the binding affinity represents another fundamental limitation. Most current scoring functions are generic models derived from large-scale experimental data of ligand-target complexes. However, a universally accurate scoring function remains the greatest challenge. Scoring functions must balance speed with accuracy, and the trade-offs inherent in this balance limit overall performance. The traditional approach to train empirical scoring functions using linear regression to optimize correlation of predicted and experimental binding affinities does not result in functions with optimal docking capabilities[18].

5.2 Computational Demands

Virtual screening of large compound libraries requires substantial computational resources. While methods like AutoDock Vina with multithreaded architecture and GPU-accelerated variants have dramatically improved throughput, screening millions of compounds remains computationally intensive. The balance between computational speed and accuracy continues to be a critical consideration in method development[19, 5].

5.3 Validation Requirements

Computational predictions must ultimately be validated through experimental testing. The gap between computational predictions and experimental validation can be substantial, with many computationally predicted hits failing to show activity in biological assays. The reasons could be several folds including the assumptions incorporated in developing scoring functions to improve computational speed. Many scoring functions entirely ignore entropy component of binding, which is significant penalty component of binding free energy (ΔG).

5.4 Solvation Effects

Solvation effects remain a major bottleneck in molecular docking and scoring because they couple tightly to both enthalpic and entropic components of binding, yet must be approximated for enhanced throughput. Most docking scoring functions treat the solvent implicitly or via empirical terms, which can reproduce experimental solvation energies in isolation but often fail to improve, and sometimes even worsen. As a result, the incomplete and approximate treatment of solvation is recognized as one of the principal reasons why current docking scores show limited quantitative reliability for affinity prediction, even when sampling is adequate[3].

6 Upcoming improvments

6.1 Artificial Intelligence and Machine Learning

The integration of artificial intelligence and machine learning with structure-based drug design represents one of the most promising future directions. AI can transform drug discovery and early drug development by addressing inefficiencies in traditional methods. Machine learning-based scoring functions are showing promise for outperforming traditional approaches while displaying greater generalizability. Geometric deep learning, an emerging concept of neural network-based machine learning, has been applied to macromolecular structures with particular relevance for structure-based drug discovery. Applications include molecular property prediction, ligand binding site and pose prediction, and structure-based de novo molecular design. Deep learning models can leverage interatomic potentials between complexes' bound and unbound states for robust predictions.

6.2 Quantum Computing

The upcoming power of quantum computing promises to enhance molecular dynamics simulations and drug discovery capabilities. Quantum computing could potentially address some of the computational limitations that currently constrain full molecular flexibility treatment and extensive conformational sampling. The powerful synergy of AI, machine learning-enhanced molecular dynamics simulations, and quantum computing will aid in developing innovative therapeutic strategies.

7 Summary

Structure-based drug design have fundamentally transformed pharmaceutical research, evolving from theoretical concepts to essential tools in modern drug discovery. The field has delivered remarkable successes, from HIV protease inhibitors to recent breakthroughs in targeting previously undruggable proteins like KRAS. The Protein Data Bank's contribution to facilitating discovery of approximately 90% of new drugs approved by the FDA underscores the profound impact of structural information on therapeutic development. Despite significant advances, challenges remain in achieving universal accuracy in binding affinity prediction, handling protein flexibility, and developing scoring functions that perform optimally across diverse target classes. The integration of artificial intelligence, machine learning, and revolutionary structure prediction methods like AlphaFold promises to address many current limitations and open new frontiers in drug discovery. The future of SBDD will likely see greater integration of multiple computational approaches, from molecular docking and dynamics simulations to machine learning and quantum computing. As computational methods become more sophisticated and experimental structural data continues to accumulate, structure-based drug design will play an increasingly central role in discovering safe and effective therapeutics. For beginners entering this field, the foundations presented in this bank provide a medium to explore deep in this area, understand the subtle nuances and make informed choices on several tools available.

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